Ethan Hosier

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Education

Imperial College London — London, UK
MEng Computing - First Class Honours — *Oct. 2022 – June 2026*

Experience

Qube Research & Technologies (QRT) | *Kotlin, Python, AWS* — Jun 2026–Present
Software Engineer, Intern

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Palantir | *Python, Spark, TypeScript* — Apr–Sep 2025
Forward Deployed Software Engineer, Intern — *London, UK*

- Parallelised a graph-based Spark pipeline processing billions of transactions/hour, reducing execution time by 71%.
- Introduced caching of intermediate results and incremental processing to avoid full graph recomputation.
- Developed a spreadsheet engine supporting real-time collaboration, formulas and keyboard shortcuts, migrating the client's financial forecasting workflow to Palantir Foundry.
- Built a model router for Palantir's internal chatbot using a lightweight classifier to route requests to configurable, task-specific models

Revolut | *Java, TypeScript, React.js, PostgreSQL, Google Cloud Platform* — Jul–Sep 2024
Software Engineer, Intern — *London, UK*

- Developed an in-process cache updated from a Postgres event stream, reducing query latency by 92%.
- Reduced production pricing-rule execution time by 24% through caching and parallelising independent calculations.
- Built a dashboard for product and pricing-rule configurations, used daily by the CEO and leadership team, eliminating the need for manual SQL queries.

Projects

PintOS Operating System Kernel | *C*

- Implemented demand paging with clock-based page replacement, swapping, stack growth and memory-mapped files
- Added argument passing and process creation/wait handling for user programs, and implemented system calls for process control and I/O
- Implemented priority-based kernel thread scheduling with priority donation and a multilevel feedback queue

Refactoring Analysis & Synthesis Tool | *Kotlin, Java, TypeScript*

- Processed IDE events into debounced edit snapshots, reconstructing refactoring sessions as replayable Git histories.
- Synthesised alternative refactoring sequences using dependency graphs and prefix-trie search.
- Achieved 83.9% category-level accuracy across four detection categories on 45 controlled evaluation sessions.

WACC Compiler | *Scala*

- Implemented WACC parsing and semantic checks in Scala, including type checking and scoped symbol resolution
- Generated x86-64 assembly for functions, arrays, pairs, and control flow, with stack and register allocation
- Implemented constant folding, function inlining, dead code elimination and peephole optimisations, reducing execution time by 12%

Technical Skills

Rust, C/C++, Kotlin, Java, Python, Go, TypeScript, AWS, React.js, Terraform